

MH1301 Discrete Mathematics

Handout 2 sup: Brief notes on conditional statement
and proof writing

Conditional Statement

Definition

Let p and q be statement variables. The **conditional** of q by p is

- “If p then q ”
- or “ p implies q ”
- denoted by $p \rightarrow q$.

It is false when p is true and q is false; otherwise it is true.

We call p the **hypothesis** (or **antecedent**) of the conditional and q the **conclusion** (or **consequent**).

Truth table

Truth table for $p \rightarrow q$:

p	q	$p \rightarrow q$
1	1	1
1	0	0
0	1	1
0	0	1

- Notice that if p is false, then $p \rightarrow q$ is always true.
- The conditional statement is said to be **vacuously true** or **true by default**.
- Example: “If I am superman, then you are spiderman.”

$$\neg p \vee q$$

Exercise: Show that $p \rightarrow q \equiv \neg p \vee q$. (Tutorial)

Application:

Example 2.2.4. Rewrite the following statement in if-then form.

Either you get to work on time or you are fired.

Solution:

- Let $\neg p$ be “You get to work on time”;
- and q be “You are fired”.
- Then p is “You do not get to work on time”.
- Therefore the statement is equivalent to
“If you do not get to work on time, then you are fired”.

Question: Is this answer acceptable?

“If you are not fired, then you got to work on time.”

Negation of a conditional statement

The negation of “if p then q ” is logically equivalent to p and not q .

$$\neg(p \rightarrow q) \equiv p \wedge \neg q.$$

We show this by using truth table or by

$$\begin{aligned}\neg(p \rightarrow q) &\equiv \neg(\neg p \vee q) \\ &\equiv \neg(\neg p) \wedge (\neg q) && \text{(by De Morgan's law)} \\ &\equiv p \wedge \neg q. && \text{(by double negative law)}\end{aligned}$$

Example

Write negations for each of the following statements.

“If you attend all lectures, then you will get A’s for all of your courses.”

Answer: You attended all lectures and you did not get A’s for all of your courses.

“If you are caught littering, then you will be fined.”

Answer: You are caught littering but you are not fined.

caution

The negation of an if-then statement **does not start with the word if.**

Contrapositive of a Conditional Statement

Definition

The **contrapositive** of a conditional statement of the form “If p then q ” is

$$\text{If } \neg q \text{ then } \neg p$$

Symbolically, the contrapositive of $p \rightarrow q$ is $\neg q \rightarrow \neg p$.

(See Exercise 2.2.26)

$$p \rightarrow q \equiv \neg q \rightarrow \neg p.$$

Getting proofs started

Identify “Starting point” and “Conclusion to be shown”.

Example. Prove that a group of order 36 is not simple.

.....

Proof:

Let G be a group of order 36. (Starting Point / hypothesis.)

$\vdots$

$\therefore G$ is not simple. (Conclusion.)

Method of proof by contradiction

$$\forall x, \text{ if } p(x) \text{ then } q(x)$$

Method of Proof by Contradiction

1. Suppose the statement to be proved is false.

That is, suppose that the negation of the statement is true. (Be very careful when writing the negation!)

2. Show that this supposition leads logically to a contradiction.
3. Conclude that the statement to be proved is true.

Suppose not. Then $\exists x$ s.t. $p(x) \wedge \neg q(x)$

Example – Theorem 4.6.1

Theorem 4.6.1

There is no greatest integer.

Proof:

1. Suppose not. Then there exists a greatest integer, say $N \in \mathbb{Z}$.
Then $\forall m \in \mathbb{Z}$, we have $m \leq N$ (since N is the greatest).
2. Take $M := N + 1$.
Then $M \in \mathbb{Z}$ and $M > N$.
3. This is a contradiction to N being the greatest integer.

Theorem 4.6.2

Theorem 4.6.2

There is no integer that is both even and odd.

Suppose not. Then $\exists N \in \mathbb{Z}$, that is both odd and even.

1. N is even, so there exists $k \in \mathbb{Z}$ such that $N = 2k$.
 N is odd, so there exists $l \in \mathbb{Z}$ such that $N = 2l + 1$.

$$\therefore 2k = 2l + 1.$$

$$\therefore 2(k - l) = 2k - 2l = 1.$$

2. $\therefore k - l = \frac{1}{2} \notin \mathbb{Z}.$

But $k - l \in \mathbb{Z}$ since it is the difference of two integers.

3. This is a contradiction to $k - l$ is an integer and $k - l$ is not an integer.

Theorem 4.6.3

Theorem 4.6.3

The sum of any rational number and any irrational number is irrational.

Suppose not. Then $\exists r \in \mathbb{R}$ and $\exists s \in \mathbb{R}$ such that $r \in \mathbb{Q}$, $s \notin \mathbb{Q}$, and $r + s \in \mathbb{Q}$.

1. Then $r = \frac{a}{b}$ for some $a \in \mathbb{Z}$ and $b \in \mathbb{Z}$ with $b \neq 0$, and $r + s = \frac{c}{d}$ for some $c \in \mathbb{Z}$ and $d \in \mathbb{Z}$ with $d \neq 0$.

$$\therefore s = (r + s) - r = \frac{c}{d} - \frac{a}{b} = \frac{bc - ad}{bd}.$$

2. Note that $bc - ad \in \mathbb{Z}$. (It is a difference of products of integers.)
Note also that $bd \neq 0$ by the zero product rule.
 $\therefore s \in \mathbb{Q}$.

3. This is a contradiction to $s \in \mathbb{Q}$ and $s \notin \mathbb{Q}$.

Method of Proof by Contraposition

1. Express the statement to be proved in the form

$$\forall x \in D, \text{ if } P(x) \text{ then } Q(x).$$

(This step may be done mentally.)

2. Rewrite this statement in the contrapositive form.

$$\forall x \in D, \text{ if } \neg Q(x) \text{ then } \neg P(x).$$

(This step may be done mentally.)

3. Prove the contrapositive statement by a direct proof.

- a. Suppose x is a (particular but arbitrarily chosen) element of D such that $Q(x)$ is false.
- b. Show that $P(x)$ is false.

Example – Proposition 4.6.4

Proposition 4.6.4

For all integers n , if n^2 is even, then n is even.

Proof:

1. (W.T.S. $\forall n \in \mathbb{Z}$, if n^2 is even, then n is even.)

2. (W.T.S. $\forall n \in \mathbb{Z}$, if n is not even, then n^2 is not even.)
(i.e. W.T.S. $\forall n \in \mathbb{Z}$, if n is odd, then n^2 is odd.)

3a. Suppose n is an odd integer.
Then $n = 2k + 1$ for some integer k .

3b. Then $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$.
 $\therefore n^2$ is an odd integer.

Proof by Contradiction and Proof by Contraposition

Exactly the same sequence of steps used in a proof by contraposition can be used to present a proof by contradiction.

W.T.S. $\forall x \in D$, if $P(x)$ then $Q(x)$.

Proof by contraposition:

- 1-2. Suppose given $x \in D$ such that $\neg Q(x)$ holds.
3. Show $\neg P(x)$.

Proof by contradiction:

1. Suppose given $x \in D$ such that $P(x)$ and $\neg Q(x)$ holds.
2. Show $P(x) \wedge \neg Q(x) \rightarrow F$.

Note: The steps in contraposition gives us $\neg P(x)$.

Use those steps in our proof by contradiction.

Therefore, we have $P(x) \wedge \neg Q(x) \wedge \neg P(x)$.

Therefore we have F .

Proposition 4.6.4

Proposition 4.6.4

For all integers n , if n^2 is even, then n is even.

Proof:

1. Suppose not. Then there exists an integer n where n^2 is even and n is not even, (i.e. n is odd.)
Since n is odd, $n = 2k + 1$ for some integer k .
2. Then $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$.
 $\therefore n^2$ is an odd integer.
3. $\therefore n^2$ is both even and odd.
This is a contradiction to Theorem 4.6.2.